

# Incompatible clauses in Conjecture 2171

Chengzhe Gao

The source asserts both that Frankl’s union-closed conjecture holds and that the ground-set bound twelve for a minimal counterexample is attained. These two clauses are incompatible. This observation neither proves nor disproves the classical Frankl conjecture considered alone.

For a finite set family  $\mathcal{F}$ , put  $d_{\mathcal{F}}(x) = |\{A \in \mathcal{F} : x \in A\}|$ . Use the ground set  $\bigcup \mathcal{F}$ , and exclude the degenerate family whose only possible member is the empty set. The usual Frankl assertion is

$$\forall \mathcal{F} \text{ union-closed, } \exists x \in \bigcup \mathcal{F} : 2d_{\mathcal{F}}(x) \geq |\mathcal{F}|.$$

A counterexample is an admissible union-closed family for which  $2d_{\mathcal{F}}(x) < |\mathcal{F}|$  for every ground element.

**Theorem 1.** *Frankl’s assertion and the existence of a minimal counterexample with ground-set size twelve cannot both hold.*

*Proof.* Assume both statements, and take the claimed family on twelve ground elements. Frankl’s assertion applies to that family and supplies an element  $x$  with  $2d_{\mathcal{F}}(x) \geq |\mathcal{F}|$ . The counterexample condition gives the strict opposite inequality for the same  $x$ , a contradiction. Minimality and the value twelve are not needed for the contradiction.  $\square$

The words “with twelve attained” in English and the corresponding Chinese phrase (“and attained”) are an existential assertion, not merely a conditional lower bound on hypothetical counterexamples. Removing attainment would remove this contradiction and would be a different problem.

**Formal verification.** Family `n` is a list of actual subsets of  $\text{Fin}(n)$ , represented by Boolean membership functions. `Admissible` requires no repeated sets, closure under pointwise union, full ground-set support, and the presence of a nonempty set. Every finite set family with an  $n$ -element ground set admits such a representation by labeling its ground elements; union and occurrence counts are preserved. Conversely each admissible representation is such a family. `frequency` computes the actual number of sets containing an element. The theorem `counterexample_iff` connects the strict inequalities to absence of a Frankl witness. `MinimalGroundCounterexample` requires an actual counterexample and the absence of any smaller-ground counterexample. The final `conjecture2171_false` negates the conjunction of the universal Frankl assertion and an actual attained twelve-element minimum. Thus the formalization does not replace the set theory by an uninterpreted proposition. It proves only the inconsistency of the source’s full statement.

Lean 4.19.0 with `Std` checks the proof, with only standard logical axioms in the printed dependency lists.